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EELGs in GMOS ZZZ

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ABSTRACT

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1. INTRODUCTION

2. DATA

The program GS-2017A-FT-9 (PI: Kiyoto Yabe) is a Fast Turnaround program that was found within the Gemini Science Archive as part of a larger search of all spectroscopic data that were taken within the COSMOS field (Khostovan et al, in prep ZZZ). The main focus of the program was to target extremely strong emission line galaxies at $z > 0.3$ that were identified by strong nebular excess in the Subaru/HSC broadband filters. These sources are expected to have strong [OIII] EW of > 2000Å and are metal-poor with the possibility of [OIII]4363Å detection.

2.1. *GMOS Spectroscopy*

GS-2017A-FT-9 consisted of three masks with only one in COSMOS which is the focus of this paper. Observations were taken in 2017 April 23 on Gemini-South using the Gemini Multi-Object Spectrograph (GMOS) with the Hamamatsu Detector, R150 grating, and GG455 blocking filter. This allows for spectral coverage from ZZZ to ZZZ with the blocking filter ensuring second-order ZZZ. Table 1 shows the observation log identifying the frame ID as found in the Gemini Science Archive, along with the frame type, exposure time, and central wavelength (dispangle). Seeing conditions were reported as 0.9″ throughout the observing time and humidity levels were somewhat elevated at ∼ 40%. A total of 3 individual exposures were pointed at GS2017AFT009-02 (mask design file) with the central wavelengths of each exposure being 8070Å, 8200Å, and 8330Å, respectively, which when 2D Coadded would take into account the GMOS detector chip gaps allowing for continuous spectral coverage in the final 1D and 2D spectra. Each science exposure was 1200 seconds resulting in an on-source integration time of 3600 seconds. Slit widths were set to 0.5″ for the science targets and box slits of 2″ sizes were used for alignment stars. Calibration frames consisted of a single 5 second twilight flat taken in the beginning of the night, a single 1 second dome flat (GCALflat) for each dispangle, and a 20 second CuAr lamp exposure for wavelngth calibration per each dispangle. Bias frames were taken on 2017 April 29 (last night of the program) and consisted of 5 frames each 1 second in exposure time.

The standard star LTT7379 (G0 star) that we use for flux calibration purposes was observed on 2017 April 29 with the same configuration and central wavelength of 8200Å. Conditions were reported with seeing as 1.08″ and humidity levels of 14%. Observations of the standard star were done using a 0.5″ wide longslit for an on-source integration time of 10 seconds. Associated CuAr lamp and Dome Flat frames were observed in 1 second exposures with the same configuration and slit widths.

2.2. *COSMOS2020 Classic*

We match sources within the COSMOS GS-2017A-FT-9 mask to COSMOS2020 Classic (Weaver et al. 2022) which has high-quality, multi-wavelength photometry enabling us to study spectral energy distributions of the targets. A total of 38 photometric bands are included from space-based missions (*GALEX*, *HST*, *Spitzer*) and ground-based obseratories (CFHT/MegaCam, Subaru/SuprimeCam, Subaru/HyperSuprimeCam, Paranal/VISTA). ZZZ

2.3. *Ancillary F140W imaging*

We also include ancillary *HST*/WFC3 F140W imaging in our analysis found within MAST archive that includes our main source of interest. The ancillary imaging is part of *HST* PID #15115 (PI: John Silverman; Silverman et al. (2018)) that targeted a sample of $z \sim 1.5$ broad-line AGN to study supermassive black hole properties and how it relates to galaxy mass. Our main source of interest was in close proximity their main target (LID360) with the F140W image having an exposure time of 2400 seconds and 3σ depth of ZZZ.

HST Program 15115 – PI: John Silverman Focused on a $z \sim 1.5$ AGN

Post in a cutout image

3. METHODOLOGY

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Table 1. Observation Log for GS-2017A-FT-9 and mask GS2017AFT009-03. Dispangle is the central wavelength of the observation. No bias frame was take on 2017 April 23 which is why we use the bias associated with this program that was taken on 2017 April 29.

Frame ID	Frame Type	Exposure Time (sec)	Dispangle (Å)
Science & Associated Calibrations			
S20170423S0039	Science	1200.0	8070
S20170423S0038	Science	1200.0	8200
S20170423S0041	Science	1200.0	8330
S20170423S0113	CuAr Lamp	20.0	8070
S20170423S0112	CuAr Lamp	20.0	8200
S20170423S0114	CuAr Lamp	20.0	8330
S20170423S0040	Dome Flat	1.0	8070
S20170423S0037	Dome Flat	1.0	8200
S20170423S0042	Dome Flat	1.0	8330
S20170423S0014	Twilight Flat	5.0	8070
S20170423S0014	Twilight Flat	5.0	8200
S20170423S0014	Twilight Flat	5.0	8330
S20170429S0198	Bias	1.0	–
S20170429S0199	Bias	1.0	–
S20170429S0200	Bias	1.0	–
S20170429S0201	Bias	1.0	–
S20170429S0202	Bias	1.0	–
Standard Star & Associated Calibrations			
S20170429S0169	Standard	10.0	8200
S20170429S0180	CuAr Lamp	1.0	8200
S20170429S0170	Dome Flat	1.0	8200
S20170429S0243	Bias	1.0	–
S20170429S0244	Bias	1.0	–
S20170429S0245	Bias	1.0	–
S20170429S0246	Bias	1.0	–
S20170429S0247	Bias	1.0	–

3.1. *Data Reduction*

We reduce all raw spectroscopic data using `Pypeit` (Prochaska et al. 2020a; `v1.10dev` – `gemini_gmos_mask_ingestion` branch), a `python` semi-automated data reduction pipeline that supports many ground- and space-based facilities. The pipeline starts by reading setup files which list the location of the science/standard & associated calibration frames. These setup files are made available via the linked github repository[1] for any one to reproduce the final spectra shown in this work. The first main script of the pipeline automatically does bias

[1] ZZZ GITHUB LINK

Table 2. Multiwavelength photometry of EELG1002 used in our SED fitting. All data are drawn from the COSMOS2020 Classic catalog (Weaver et al. 2022) except for the CFHT/WIRCam (from COSMOS2015; Laigle et al. 2016) and *HST*/WFC3 F140W (see §2.3). Auto magnitudes are Both *GALEX*/FUV and NUV photometry does exist for EELG1002 and is available in the COSMOS2020 catalog; however, it is affected by blending issues with 2 nearby sources. Same is true for *Spitzer*/IRAC photometry. We ignore those measurements in our SED fitting process.

Band	λ_c (Å)	FWHM (Å)	Magnitude (AB mag)
CFHT/MegaCam			
u^*			23.93 ± 0.01
u			23.84 ± 0.01
Subaru/HyperSuprimeCam			
g			23.84 ± 0.01
r			23.82 ± 0.01
i			23.82 ± 0.01
z			22.12 ± 0.01
y			24.20 ± 0.04
Subaru/SuprimeCam			
B			24.06 ± 0.01
g^+			23.87 ± 0.01
V			23.78 ± 0.02
r^+			23.84 ± 0.02
i^+			23.84 ± 0.03
z^+			22.56 ± 0.02
z^{++}			22.54 ± 0.01
$IB427$			23.93 ± 0.04
$IB464$			23.76 ± 0.04
$IA484$			23.99 ± 0.04
$IB505$			23.78 ± 0.04
$IA527$			23.93 ± 0.03
$IB574$			24.00 ± 0.05
$IA624$			23.76 ± 0.04
$IA679$			23.43 ± 0.03
$IB709$			23.68 ± 0.03
$IA738$			23.83 ± 0.05
$IA767$			24.38 ± 0.08
$IA827$			24.31 ± 0.07
$NB711$			23.26 ± 0.06
$NB816$			24.11 ± 0.06
HST			
ACS/F814W			23.37 ± 0.01
WFC3/F140W			23.89 ± 0.08
CFHT/WIRCam			
H			23.82 ± 0.26
K_s			24.39 ± 0.44

& overscan subtraction, slit edge detections, flat fielding, removal of cosmic rays, wavelength calibrations, tilt corrections, sky subtraction, flexure corrections, and a first pass on object extraction using the optimal extraction Horne (1986) and boxcar approach.

At the time that data was reduced, `Pypeit` officially supported the use of slit mask designs for Keck instruments to identify slit edges and object positions along the sit. This was not officially supported for Gemini/GMOS but was available in the `gemini_gmos_mask_ingestion` development branch which is why we reduced all our data using this specific branch. Currently this has now been merged into the official `Pypeit` release versions.

The second step in the pipeline is to 2D coadd all 3 reduced science frames of varying central wavelengths (accounts for GMOS chip gaps in final 2D coadded spectra). To account for potential drift during observations, we define the offsets by using the brightest object in the mask. `Pypeit` models the spatial profile of the object and calculates the spatial shift in reference to the first science frame in our 2D coadd list. These were 0.05 and 0.56 pixels for science frames 38 and 41, respectively (check UNITS ZZZ). This is then stacked with weights corresponding to the S/N of our brightest object.

3.1.1. *Flux Calibration & Slitloss Corrections*

Flux calibration is done by first reducing our standard star LTT7379 (G0) using the same `Pypeit` reduction as mentioned above. We then use the wavelength-calibrated, reduced spectra to measure the sensitivity curve that converts raw counts into physical flux units. `Pypeit` does this by comparing the observed 1D GMOS spectra against archival spectra drawn, in this case, from ESO Optical and UV Spectrophotometric Standard Stars library. The observed resolving power ($R \sim 2100$) is also provided in the sensitivity function configuration file which `Pypeit` uses to match the archived LTT7379 spectra to the same resolution as our GMOS observation. The final sensitivity curve is then applied to each of our science targets.

An additional correction is also necessary to account for aperture/slitloss where a percentage of the total flux of an object falls outsides of the slit resulting in a reduction in observed flux that can not be corrected with the flux calibration approach listed above. As such, we apply a slitloss correction using *HST* ACS/F814W imaging of our targets as a starting point taking advantage of the high spatial resolution ($\sim 0.03''$/pixel). We then smooth out the F814W image using a Gaussian Kernel set to a FWHM of ~ZZZ (seeing) to best match the ground-based conditions at the time of observations. The smoothed F814W image is then used to calculate the total flux of our main targets within a slit with $0.5''$ width. This is compared to the total F814W flux which gives a percentage of the total flux that fell within the slit. The remaining fraction of flux missing is then accounted for by simply multiplying the 1D flux-calibrated spectra $1 + f$ with $f = 0.456$ being the ratio of missing-to-total flux.

3.2. *Spectroscopic Redshift & Emission Line Profile Fitting*

We use `PyQSOFit` (Guo et al. 2018) for fitting the profiles of all emission lines observed in the GMOS spectra which requires an initial estimate of the spectroscopic redshift visually measured as $z = 0.8275$ using SpecPro SPECPRO PAPER. Each detected emission line is fit by a single Gaussian profile centered on the expected rest-frame wavelength assuming our initial redshift. The center of each Gaussian is allowed to vary up to ~ 2.7Å allowing for variations in the final measured spectroscopic redshift. Table 3 shows the best-fit redshifts about the line centers consistent with $z = 0.8275^{+ZZZ}_{-ZZZ}$.

Line widths of each emission line is set to be constant among other associated lines within the atomic specie (e.g., Balmer line FWHMs are the same from Hβ to Hη). This allows us to deblend lines from different atomic species that are in close wavelength proximity to one another (e.g., [NeIII]3968Å and Hϵ separated by 2.6Å; rest-frame) and are not fully resolved given the GMOS wavelength resolution limit. The [OII]3726,3729Å doublet is also not resolved for which we fit two Gaussians assuming the FWHM from [OIII]5007Å to deblend the doublet which is important to measuring electron density. Line fluxes are measured by integrating the best-fit Gaussian profiles.

Uncertainty measurements in the emission line profiles are measured via Markov Chain Monte Carlo (MCMC) where `PyQSOFit` samples the error spectrum a total of 1000 times and perturbs the 1D spectra accordingly in each realization. We ignore the first 200 samples in the MCMC (20% burn-in) which is done to ensure that starting points do not affect the underlying probability distribution functions and uncertainty measurements. Overall, each emission line detected has an associated probability distribution that will be used throughout this work to measure line ratios and associated uncertainties.

3.3. *Measuring ISM Properties*

In this section, we will describe how we use nebular emission line ratios in conjunction with `PyNeb` (Luridiana et al. 2015) to measure key ISM properties. Below are the main key definitions and conventions that we adopt in defining key line ratios:

- O3HB: [OIII]5007Å/Hβ
- O32: [OIII]5007Å/[OII]3726, 3729Å
- Ne3O2: [NeIII]3869Å/[OII]3726, 3729Å
- Ne3O3: [NeIII]3869Å/[OIII]5007Å
- $R2$: [OII]/Hβ
- $R3$: ([OIII]5007Å + [OIII]4959Å)/Hβ
- $R23$: ([OIII]5007Å + [OIII]4959Å + [OII])/Hβ

3.3.1. *Dust Extinction*

Balmer decrements (e.g., Hα/Hβ) trace the level of dust extinction within HII regions and are used to dust-correct observed line fluxes when coupled with an assumed dust attenuation curve. The reddening is measured by comparing the observed-to-intrinsic Balmer ratios as shown below:

$$E(B-V) = \frac{2.5}{k(\mathrm{H}\gamma) - k(\mathrm{H}\beta)} \log_{10}\left(\frac{[\mathrm{H}\beta/\mathrm{H}\gamma]_{\mathrm{obs}}}{[\mathrm{H}\beta/\mathrm{H}\gamma]_{\mathrm{int}}}\right) \quad (1)$$

coupled with the dust attenuation curve evaluated at the wavelength of each Balmer line used, $k(\lambda)$, which we assume the Calzetti et al. (2000) attenuation curve. Amount of dust extinction is then measured as $A_\lambda = k(\lambda)E(B-V)$ at any given wavelength. We use both the Hβ/Hγ and Hβ/Hδ ratios observed in the GMOS spectra to measure $E(B-V)$ and A_λ and use `PyNeb` to determine the intrinsic line ratios, Hβ/Hγ = 2.11 and Hβ/Hδ = 3.82, assuming case B recombination, T_e = 15000 K (electron temperature), and $n_e = 100\ \mathrm{cm}^{-3}$ (electron density). Varying these assumptions marginally changes the intrinsic ratios (e.g., Hβ/Hδ = 3.81 for $n_e = 1000\ \mathrm{cm}^{-3}$ and T_e = 15000 K) and does not affect our dust extinction measurements. Uncertainties in $E(B-V)$ are measured via each MCMC realization used in the emission line profile fitting (see §3.2).

3.3.2. *Electron Temperature & Density*

The electron temperature is measured using a two-zone HII region model due to [OIII] (doubly ionized oxygen) tracing the (high ionizing) zone closest to the ionizing source and [OII] (single ionized oxygen) being able to trace the outer (low ionizing) zone of an HII region. Electron temperature traced via [OIII] , $T_e(\mathrm{O}^{++})$, is measured using the auroral [OIII]4363/[OIII]5007Å line ratio at a fixed $n_e = 1000\ \mathrm{cm}^{-3}$ and the `getTemDen` function in `PyNeb`. The fixed n_e is done arbitrarily as T_e is essentially independent of electron density up to $\sim 10^4\ \mathrm{cm}^{-3}$ (which is not expected for the ISMs of galaxy populations) but is an input in `getTemDen`.

Tracing the low ionizing zone electron temperature requires O^+ auroral lines (e.g., [OII]7322,7332Å) which is outside our GMOS coverage. Instead, we estimate O^+ electron temperature using the Izotov et al. (2006) calibration:

$$t_e(\mathrm{O}^+) = t_e(\mathrm{O}^{++}) \times (2.065 - 0.498 t_e(\mathrm{O}^{++})) - 0.577 \quad (2)$$

where $T_e(\mathrm{O}^+) = t_e(\mathrm{O}^+) \times 10^4$ K and $T_e(\mathrm{O}^{++}) = t_e(\mathrm{O}^{++}) \times 10^4$ K. We note that this is the low metallicity calibration which we later show in this paper to be appropriate for EELG1002.

We then measure the electron density using the [OII]3726,3729Å doublet which is sensitive to T_e. For example, n_e can vary from $\sim$ 500 cm^{-3} to $\sim$ 640 cm^{-3} by increasing T_e from 10000 K to 20000 K assuming the [OII] doublet ratio is of order unity. Therefore, it is crucial that we first measure the electron temperature as described above and then measure electron density. We use `PyNeb getTemDen` function with the estimated $T_e(\mathrm{O}^+)$ and the observed [OII]3729/[OII]3726Å ratio.

3.3.3. *Gas-Phase Abundance*

Abundances associated with Oxygen and Neon are measured using `PyNeb` and its `getIonAbundance` feature assuming the electron temperatures in both the high and low ionizing zones. Our approach also assumes that the electron density as measured from [OII] is constant throughout the ionizing zones. Below we describe how the ionic and total abundances are measured.

Oxygen abundance is measured with the formalism:

$$\frac{\mathrm{O}}{\mathrm{H}} \approx \frac{\mathrm{O}^+}{\mathrm{H}} + \frac{\mathrm{O}^{++}}{\mathrm{H}} \quad (3)$$

where additional abundance contribution from O^{+++} is not expected to contribute significantly to the overall O/H abundance. Specifically, to have O^{+++} requires 54.9 eV and, therefore, we would also expect Helium recombination line detections within the GMOS spectra (e.g., HeI4686Å) for which we find none. Berg et al. (2018) investigated $z \sim 2$ lensed source with OIV detection and found that $\mathrm{O}^{+++}/\mathrm{H}$ contribution only accounted for a $\sim$ 5% increase (+0.02 dex) in the total O/H abundance. We therefore continue measuring O/H abundance as defined in Equation 3. The $\mathrm{O}^{++}/\mathrm{H}$ abundance is measured using $R3$ with n_e and $T_e(\mathrm{O}^{++})$ measured in §3.3.2 and O^+/H is measured using $R2$, n_e, and $T_e(\mathrm{O}^+)$.

Figure **??** also shows clear detections of both the [NeIII]3869Å and [NeIII]3968Å lines, although the latter is affected by Hϵ contribution. Therefore, we can also investigate the Ne/H abundance, as well as Ne/O abundance ratios. We define the Neon-to-Hydrogen abundance as:

$$\frac{\mathrm{Ne}}{\mathrm{H}} \approx \frac{\mathrm{Ne}^{++}}{\mathrm{H}} \times \mathrm{ICF} \quad (4)$$

where the ionization contribution fraction (ICF) traces the relative contribution from other Neon ionization states (e.g., Ne^+). We adopt the ICF formalism of Izotov et al. (2006):

$$\mathrm{ICF} = 1.365 - 0.385\frac{\mathrm{O}^{++}}{\mathrm{O}^+ + \mathrm{O}^{++}} + 0.022\frac{\mathrm{O}^+ + \mathrm{O}^{++}}{\mathrm{O}^{++}} \quad (5)$$

which assumes a low metallicity system (consistent with EELG1002; see §ZZZ). The $\mathrm{Ne}^{++}/\mathrm{H}$ abundance is measured using the [NeIII]3869Å/Hβ ratio along with our measured n_e and $T_e(\mathrm{O}^{++})$ as [NeIII] requires 41 eV comparable to the 35 eV needed for [OIII] emission.

3.3.4. *Ionization Parameter*

The ionization parameter is defined as:

$$U = \frac{Q_{H^0}}{4\pi R_S^2 c n_H} \quad (6)$$

where Q_H^0 is the number of hydrogen ionizing photons per unit time, R_S is the Strömgen radius, c is the speed of light, and n_H is the hydrogen number density. Essentially it traces how many ionizing photons are produced within an HII region in comparison to the present number of hydrogen atoms.

The higher the ionization parameter would signify more ionizing photons and/or less hydrogen atoms that can be interpreted as energetic ISM conditions caused by: harder ionizing radiation field (ZZZ), electron density (ZZZ), higher SFR (ZZZ), lower metallicities (ZZZ), and higher SFR surface densities (e.g., ZZZ NAVEEN).

The classical diagnostic used to trace U is the O32 ratio given that O^{++} requires 35 eV compared to O^{+} which requires 13.6 eV such that it is sensitive to the ionization state of gas within the HII region (ZZZ). Another diagnostic typically used is the Ne3O2 ratio where Ne^{++} requires $\sim$ 41 eV and is insensitive to dust attenuation given the $\sim$ 140Å separation between the two lines (Levesque & Richardson 2014). However, such diagnostics are limited in range of U and gas-phase metallicities (Kewley et al. 2019). Therefore, we measure ionization parameters using three methods that include photoionization modeling with U as an input: HCM (Pérez-Montero 2014), BAGPIPES (Carnall et al. 2018), and CIGALE (Boquien et al. 2019).

We use HCM v5.3 (latest version) which uses CLOUDY v17 (ZZZFerland2017) with a fixed $n_e = 100$ cm^{-3} to model the nebular gas. This is coupled with BPASS v2.1 (ZZZEldrige2017) that assumes an instantaneous burst of star formation with an age of 1 Myr, IMF with slope -1.35, upper mass limit of 300 $M_\odot$, and includes binary stellar populations. The combination of both forms a grid of modeled line ratios (e.g., [OIII]/[OII]) as a function of O/H, N/O, and U. The observed line fluxes and ratio are then compared to the modeled grid and used to measure the best-fit gas-phase abundances and U. We refer the reader to §3.4.1 and 3.4.2 for details on how CIGALE and BAGPIPES, respectively, is used to measure U via simultaneously fitting the observed GMOS 1D spectra and multi-wavelength photometry.

3.4. *Spectral Energy Distribution Fitting*

In this section, we describe how spectrophotometric spectral energy distribution (SED) fitting was done using two main codes: CIGALE and BAGPIPES where GMOS spectra and all ancillary multi-wavelength photometry described in §2 are fitted simultaneously. We do note that GALEX/FUV and NUV and *Spitzer*/IRAC photometry are not included in the SED fitting process given contamination due to blending issues by 2 nearby sources (FIGZZZ). Lastly, all photometry used in the SED fitting process have an additional 10% uncertainity included in quadrature to account for underestimated errors.

3.4.1. CIGALE

We use the latest version of `Cigale` (Boquien et al. 2019; Yang et al. 2022), a template grid-based SED fitting code, and provide a brief overview of assumptions made in the SED fitting process which are highlighted in Table 6. Setup files are also made available within the GITHUB repository [2]. Briefly, we assume a Bruzual & Charlot (2003) stellar population synthesis model coupled with a Chabrier (2003) IMF and consider stellar metallicities within the range of 0.005 and 1 $Z_\odot$. Nebular emission is modeled using the **?** templates which were generated using CLOUDY v13.01 (ZZZFerland1998,2013) with U considered in the range of -2.6 to -1.0. Gas-phase metallicity is fixed given ZZZ. CIGALE only allows for three different fixed values for n_e for which we assume 100 and 1000 cm^{-3} given n_e = ZZZ as measured in §3.3.2. Given that GALEX photometry is contaminated by 2 nearby sources, we have no constraint on potential LyC emission and therefore keep both f_{esc} (LyC escape fraction) and f_{dust} (fraction of LyC photons absorbed by dust) fixed to 0%. All emission lines in the model have fixed FWHM to 370 km s^{-1}, consistent with the observed FWHM of each line (ZZZTABLE). Dust attenuation in both stellar and nebular reddening are evaluated in the range of $E(B-V) = 0.0 - 0.15$ mag with a stellar-nebular reddening ratio order unity; however, we note that Balmer Decrements suggest no dust attenuation in EELG1002 at least within HII regions.

We assume a delayed-τ star formation history with a burst described as

$$\mathrm{SFR}(t) \propto t/\tau_{main}^2 \exp^{-t/\tau_{main}} + \exp^{-(t-t_0)/\tau_{burst}} \quad (7)$$

where SFR(t) is the star-formation rate at a given time starting at formation time, τ_{main} and τ_{burst} are the e-folding times of the old and young stellar populations, respectively, where t_0 describes the onset of bursty star formation activity. For the main stellar population (first term), we consider $\tau_{main} \sim$ 50 Myr to 7.5 Gyr and ages of 50 Myr to 6.6 Gyr. The upper range for main population age is defined as the age of the Universe at z_{spec} = ZZZ and $\tau_{main} > 6.6$ Gyr would describe a continuously rising SFH. The late burst is considered with $\tau_{burst} \sim$ 1 to 100 Myr and ages of $\sim$ 1 to 50 Myr. The total mass contribution of the late burst is defined by f_{burst} which we consider between 0 to 99%.

3.4.2. *Bagpipes*

BAGPIPES is a SED suite that allows for the simultaneous fitting of 1D spectra and multi-wavelength photometry as well as parametric and nonparametric star formation history (SFH) modeling. It uses bayesian inference along with nested sampling of the parameter space using MULTINEST to measure best-fit properties. Table ZZZ highlights the range in parameter space we allow for BAGPIPES to explore and briefly describe them here. We use the BPASS v2.2.1 SED templates (Stanway & Eldridge 2018) which includes the contribution of binary stellar populations. This assumes a broken power law IMF with a 0.1 – 0.5 $M_\odot$ slope of -1.35 followed by a steeper IMF slope of -2.35 up to an upper mass limit of 300 $M_\odot$. The broken power law shape allows for an increased contribution of older, low-mass stellar populations compared to the Chabrier (2003) IMF which exponentially cuts off at 0.1 to 1 $M_\odot$. We assume metallicities from 0.0001 to 5 $Z_\odot$ that is sampled using a logarithmic prior. The Calzetti et al. (2000) dust attenuation curve is assumed with A_V be-

[2] HTML LINK ZZZ

tween 0 to 10 mag and the ratio of nebular-to-stellar reddening set to unity. Templates for the nebular component is recomputed using the BPASS stellar grid and CLOUDY v17.03 to cover the ionization parameter range of $\log_{10} U = -4$ to -1 in order to consider cases of extreme energetic ISM conditions.

We use the Leja et al. (2019) continuity non-parametric SFH implementation in BAGPIPES to better understand how past star-formation activity could have occurred in EELG1002 based on the available spectrophotometric data. The time bins assumed in the SFH modeling starts with short duration and increases with each successive bin: 0, 1.5, 3, 10, 30, 100, 300, 1000, 3000, and 6000 Myr. The shorter time scales are to gauge how rapidly the recent burst of star formation formed and is constrained primarily by the multiple strong Balmer line detections that directly trace instantaneous star formation activity. The longer timescales are primarily constrained by photometry that well constrains the stellar continuum and is sensitive to past star-formation activity that occurred at older lookback times.

3.5. *Measuring Equivalent Width*

Equivalent width (EW) is measured as the ratio between emission line flux and its associated continuum flux density. We measure the emission line flux using the best-fit emission line profile as described in §3.2. The continuum flux density is drawn from the best-fit SEDs as measured by both CIGALE and BAGPIPES. Although a stellar-nebular reddening ratio of order unity is assumed in the SED fitting process, we note that Balmer Decrements suggest an $E(B-V)_{nebular} \sim 0$ mag while both CIGALE and BAGPIPES suggest slight stellar reddening. To account for this attenuation difference, we only correct the best-fit observed continuum flux densities using the CIGALE-measured $E(B-V)_{stellar}$ along with a Calzetti et al. (2000) dust attenuation curve and compute the equivalent widths for each line. We do not make any dust correction for the nebular emission given that our Balmer Decrement measurements show $E(B-V)_{nebular} \sim 0$ mag.

3.6. *Ionizing Photon Production Efficiency*

The ionizing photon production efficiency, ξ_{ion}, traces how well a galaxy can produce ionizing photons (e.g., > 13.6 eV that can ionize HI) and is defined as:

$$\xi_{ion} = \frac{N(H^0)}{L_{UV}^{int}} \tag{8}$$

where $N(H^0)$ represents the production rate of hydrogen ionizing photons and L_{UV}^{int} is the intrinsic, rest-frame, dust-corrected 1500Å continuum luminosity. The definition of ξ_{ion} is slightly varied where we refer the reader to §3.2 of Chevallard et al. (2018) for a detailed discussion. Some studies use the monochromatic UV luminosity attributed only to the stellar continuum (typically cited as $\xi_{ion}^{\star}$) and ignores the emission and absorption caused by neutral and ionized gas. Other studies incorporate the nebular and stellar continuum in measuring the 1500Å monochromatic luminosity ($L_{UV}^{\rm HII}$) and measure the ionizing photon production efficiency ($\xi_{ion}^{\rm HII}$). We adopt this definition throughout this paper using ξ_{ion} as our defined notation for the nebular + stellar continuum contributions.

We use the best-fit combined stellar and nebular continuum fits from CIGALE to measure L_{UV} using a tophat filter centered at 1500Å with a width of 100Å. We use the Leitherer & Heckman (1995) calibration:

$$N(H^0) = 2.1 \times 10^{12} L_{int}(\mathrm{H}\beta)\ [\mathrm{s}^{-1}] \tag{9}$$

to measure the ionizing photon production rate, $N(H^0)$, with dust-corrected Hβ luminosity from our observed GMOS spectra. Given that Balmer Decrements show $E(B-V)_{nebular} \sim 0$ mag, we do not apply any dust correction to the Hβ luminosity. However, our best-fit CIGALE measurements find $E(B-V)_{stellar} \sim$ ZZZ mag such that we need to apply a dust-correction to our L_{UV} measurement. This is done by assuming a Calzetti et al. (2000) dust attenuation curve with $A_{1500\text{Å}} =$ ZZZ mag.

4. RESULTS

4.1. *Record Breaking Low Mass, Compact, High EW Emitter at $z \sim 0.8$*

Main points: EW of this source is ZZZWhat is the stellar mass. What is the effective radius. figure showing the EW – stellar mass anticorrelation highlight that this source is quite extreme compared to typical [OIII] emitters at the same redshift and mass consistent with low-z LyC leakers even higher than EIGER measurements compared to other high EW emitters spectra doesn't have any evidence for Helium emission as expected in other sources (papers?)

4.2. *No Evidence of AGN component*

Main points: Spectra does not show broad emission line component Lack of strong ionization lines such as NeV No Chandra and XMM detections Maybe *spitzer* but source is convolved with nearby objects

MEx diagram Figure. Source falls well within the same range as SF-dominated sources BPT diagram Figure (extrapolated spectra). Source falls within the SF-dominated range and consistent with EMPGs

4.3. *Extremely Metal Poor System Consistent with $z \gtrsim 5$ Galaxies*

Main Points: Mention what the metallicity is for this object via T_e method Mass – Metallicity Figure Highlight how much this source is below the $z \sim 0.8$ relation Highlight how this matches EIGER and Langeroodi at $z \sim 8$ Consistent with EMPGs but at higher redshift

4.4. *Energetic ISM Conditions: Harder Ionizing Radiation Field*

Main Points: Measured the ionization parameter using Ne3O2, O32, Cloudy+BC03, and Cloudy+BPASS Ne3O2 gives the lowest logU and may be due to calibrations not

ISM Property	Measurement
Electron Temperature & Density	
$T_e(\mathrm{O}^{++})$	19595^{+1550}_{-1498}
$T_e(\mathrm{O}^{+})$	15544^{+83}_{-273}
n_e	458^{+356}_{-244}
Oxygen Abundance	
$12 + \log_{10}(\mathrm{O}^{++}/\mathrm{H})$	$7.451^{+0.077}_{-0.072}$
$12 + \log_{10}(\mathrm{O}^{+}/\mathrm{H})$	$6.545^{+0.044}_{-0.038}$
$12 + \log_{10}(\mathrm{O}/\mathrm{H})$	$7.501^{+0.072}_{-0.065}$
Neon Abundance	
$12 + \log_{10}(\mathrm{Ne}^{++}/\mathrm{H})$	$6.882^{+0.090}_{-0.079}$
$12 + \log_{10}(\mathrm{Ne}/\mathrm{H})$	$6.901^{+0.089}_{-0.076}$
$\mathrm{ICF}(\mathrm{Ne}^{++})$	$1.047^{+0.007}_{-0.005}$
$\log_{10}(\mathrm{Ne}/\mathrm{O})$	$-0.599^{+0.029}_{-0.025}$
Ionization Parameter – $\log_{10} U$	
`HCM+BPASS`	-1.69 ± 0.12
`Bagpipes` + `BPASS`	$-1.63^{+0.12}_{-0.13}$
`Cigale` + `BC03`	-2.27 ± 0.05
Balmer Decrement	
$\mathrm{H}\beta/\mathrm{H}\gamma$	$1.961^{+0.153}_{-0.147}$
$\mathrm{H}\beta/\mathrm{H}\delta$	$4.708^{+0.510}_{-0.432}$
$\mathrm{H}\beta/\mathrm{H}\epsilon$	$8.627^{+1.753}_{-1.323}$
$E(B-V) - (\mathrm{H}\beta/\mathrm{H}\gamma)$	$0.00^{+0.01}_{-0.00}$ mag
	0.17 mag ($< 2\sigma$)
$E(B-V) - (\mathrm{H}\beta/\mathrm{H}\delta)$	$0.30^{+0.14}_{-0.13}$ mag
Dust-Corrected Line ratios	
$[\mathrm{OIII}]_{5007\text{Å}}/\mathrm{H}\beta$	$4.899^{+0.254}_{-0.236}$
$[\mathrm{OIII}]_{5007\text{Å}}/[\mathrm{OII}]_{3726,3729\text{Å}}$	$11.428^{+0.591}_{-0.672}$
$[\mathrm{OII}]_{3726,3729\text{Å}}/\mathrm{H}\beta$	$0.431^{+0.032}_{-0.028}$
$([\mathrm{OIII}]_{5007,4959\text{Å}}/\mathrm{H}\beta$	$6.520^{+0.341}_{-0.329}$
$([\mathrm{OIII}]_{5007,4959\text{Å}} + [\mathrm{OII}]_{3726,3729\text{Å}})/\mathrm{H}\beta$	$6.948^{+0.362}_{-0.342}$
$[\mathrm{OIII}]_{4363\text{Å}}/[\mathrm{OIII}]_{5007\text{Å}}$	$0.033^{+0.004}_{-0.004}$
$[\mathrm{NeIII}]_{3869\text{Å}}/[\mathrm{OII}]_{3726,3729\text{Å}}$	$1.350^{+0.092}_{-0.082}$
$[\mathrm{NeIII}]_{3869\text{Å}}/[\mathrm{OIII}]5007\text{Å}$	$0.118^{+0.007}_{-0.006}$

compatible with low-metallicity systems BAGPIPES and HCM both use BPASS with Cloudy and find high log U

Comparison of Ne3O3 at given log U? or Stellar Mass? Maybe it is higher suggesting harder ionization field?

4.5. *Rapid Burst of Star Formation*

Main Point: Parametric SFH with Cigale suggests burst of star formation sSFR from Cigale suggests rapid mass doubling timescales Non-parametric SFH with BAGPIPES. Suggests rapid burst of star formation within the last ZZZ Myrs. Outshining could be a factor but analogs of such sources are found within Illustris. Suggests such SFHs exist and may resemble galaxies in their infancies?

Illustris Analog Search: Looked for EELG1002 analogs within the Illustris TNG simulations. Limit the search to galaxies with stellar masses in a range between 10^7 to $10^{8.7}$ $\mathrm{M}_\odot$, instantaneous SFR > 3 $\mathrm{M}_\odot$ yr^{-1}, and absolute magnitudes (`stellarphotometrics`) within the range of −18.5 and −21 magnitudes (consistent with the range of EELG1002). The wide range is to ensure we have sizable number of potential analogs to study. Vast majority of potential analogs have high sSFR but no progenitor, which could suggest either that the galaxy formed recently or that the stellar particles associated with this galaxy have recently gone above the limiting simulation resolution. In such case, we consider only those sources for which have progenitors.

Here we will want to include a SFH plot with our source of interest along with Illustris extended to z = 0. We also want to include plots of the

4.6. *Devoid of Dust or Destruction of Dust Grains?*

Main Point: Balmer Decrement suggests that E(B-V) is 0. This means there is no dust in the system But both Cigale and Bagpipes suggest some dust content which would signify some past star-formation activity and that the stellar reddening does exist. Could be that dust grains are being destroyed from intense radiation emanating from HII regions but are still found within the diffuse ISM. One way to test this is by follow-up mid IR imaging with *JWST* to observe PAH emission. Some studies already suggest that PAH is destroyed in starburst galaxies.

4.7. *High Neon Abundance*

5. DISCUSSION

Here would be a great place to talk about how this relates to high-z galaxies seen with *JWST*

5.1. *What is driving the intense star formation activity?*

Elevated Cold Gas Content? Illustris-TNG analogs of source are showing high gas masses at the level of $\sim 10^{10}$ $\mathrm{M}_\odot$! This means that the galaxy has a high gas content. Dark matter halo mass of analogs is also high such that it could be influenced by enhanced cold gas accretion. However, SFR is not high until the final snapshot. Something happens then!

In a filament? no evidence based on available data

High mass dark matter halo? Refer to my 2018 paper to get DMH mass.

5.2. *What are the potential descendants of EELG1002?*

Plots that will go here are sizes, gas mass, stellar mass evolution

5.3. *Cause of High Electron Temperatures and Ionization Parameter*

Could be due to a multitude of combined factors. Metal coolants are needed to absorb ionizing photons that heat up

Table 3. Emission Line Fits

Line	z_{spec}	Observed Line Flux (10^{-17} erg s^{-1} cm^{-2})	EW$_0$ (Cigale) (10^{-17} erg s^{-1} cm^{-2})	EW$_0$ (Bagpipes) (Å)	FWHM (Å)	σ_d (km s^{-1})
Balmer Lines						
Hβ	$0.8273^{+0.0058}_{-0.0058}$	$18.84^{+0.96}_{-0.91}$	$451.45^{+22.97}_{-21.87}$	$321.43^{+16.35}_{-15.57}$	452^{+17}_{-15}	191^{+7}_{-7}
Hγ	—	$9.61^{+0.53}_{-0.52}$	$134.48^{+7.41}_{-7.27}$	$125.97^{+6.94}_{-6.81}$	—	—
Hδ	—	$4.00^{+0.33}_{-0.34}$	$68.11^{+5.70}_{-5.72}$	$45.56^{+3.81}_{-3.83}$	—	—
Hϵ	—	$2.19^{+0.35}_{-0.36}$	$34.59^{+5.59}_{-5.69}$	$23.16^{+3.74}_{-3.81}$	—	—
Hζ+HeI3889Å	—	$0.58^{+0.26}_{-0.24}$	$8.37^{+3.80}_{-3.50}$	$5.59^{+2.54}_{-2.34}$	—	—
Hη	—	$2.90^{+0.32}_{-0.29}$	$43.46^{+4.78}_{-4.34}$	$29.04^{+3.20}_{-2.90}$	—	—
Oxygen Lines						
[OIII]5007Å	$0.8276^{+0.0058}_{-0.0058}$	$92.35^{+0.68}_{-0.70}$	$2346.39^{+17.15}_{-17.68}$	$1647.37^{+12.04}_{-12.41}$	347^{+5}_{-4}	146^{+1}_{-1}
[OIII]4959Å	—	$30.59^{+0.42}_{-0.42}$	$755.44^{+10.29}_{-10.46}$	$527.19^{+7.18}_{-7.30}$	—	—
[OIII]4363Å	—	$2.99^{+0.37}_{-0.38}$	$55.31^{+6.92}_{-7.03}$	$40.53^{+5.07}_{-5.15}$	—	—
[OII]3726,3729Å	—	$8.00^{+0.39}_{-0.36}$	$112.05^{+5.49}_{-5.01}$	$86.59^{+4.24}_{-3.87}$	—	—
[OII]3726Å	—	$3.86^{+0.38}_{-0.38}$	$53.83^{+5.32}_{-5.33}$	$44.00^{+4.35}_{-4.36}$	—	—
[OII]3729Å	—	$4.18^{+0.36}_{-0.40}$	$54.96^{+4.69}_{-5.23}$	$40.51^{+3.46}_{-3.86}$	—	—
Neon Lines						
[NeIII]3869Å	$0.8275^{+0.0058}_{-0.0058}$	$10.82^{+0.51}_{-0.47}$	$151.78^{+7.14}_{-6.61}$	$105.71^{+4.97}_{-4.60}$	538^{+28}_{-26}	227^{+12}_{-11}
[NeIII]3968Å	—	$3.35^{+0.00}_{-0.00}$	$52.89^{+0.00}_{-0.00}$	$35.48^{+0.00}_{-0.00}$	—	—

Table 4. Best-Fit SED-dervied Properties from Delayed-τ Model

Property	CIGALE
Stellar Mass (M$_\odot$)	$7.02 \pm 4.08 \times 10^7$
SFR(inst) (M$_\odot$ yr^{-1})	36.35 ± 6.63
SFR(10 Myrs) (M$_\odot$ yr^{-1})	3.82 ± 0.44
SFR(100 Myrs) (M$_\odot$ yr^{-1})	0.47 ± 0.17
f_{burst}	0.57 ± 0.26
$Z_\star$ ($Z_\odot$)	0.50 ± 0.21
$E(B-V)$	0.097 ± 0.032
sSFR(inst)	
sSFR(10 Myrs)	
sSFR(100 Myrs)	
Reduced χ^2	0.70

Table 5. Nebular AGN Diagnostics

Diagnostic	Ratio	Classification
BPT	(R3 = , N2 =)	SFG
MEx	(R3 = , Mass =)	SFG
OHN(e)O	(R3 = , NeO2 =)	ZZZ

the surrounding gas. In the case of low metallicity, there is not enough absorbants to cool the system. Couple that with low dust, this allows for gas heating and ionizing photon escape into the ISM. Indeed, this galaxy also has an inferred Lyman Continuum f_{esc} of 10% which would also decrease the ionizing parameter $\log_{10} U$.

5.4. *Ideal Case of what First Generation of Galaxies may look like?*

6. CONCLUSIONS

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This research made use of `PypeIt`[3], a Python package for semi-automated reduction of astronomical slit-based spectroscopy (Prochaska et al. 2020a,b).

Software: ASTROPY, NUMPY, PYQSOFIT, CIGALE, BAGPIPES

DATA AVAILABILITY

All raw data is publicly available and can be found within the Gemini Science Archive by searching for GS-2017A-FT-9 under Program ID. The associated pypeit reduction files can be found at githublink ZZZ and is available for public use.

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Table 6. CIGALE parameters used in the SED fitting process

Parameter	Symbol	Value
Delayed Exponential Star Formation History with Burst		
e–folding time of the main stellar population model (Gyr)	$\tau_{\rm main}$	0.05, 0.1, 0.25, 0.5, 0.75, 1.0, 2.5, 5.0, 7.5
Age of the main stellar population in the galaxy (Gyr)	$t_{\rm main}$	0.05, 0.1, 0.25, 0.5, 0.75, 1.0, 2.5, 5.0, 6.6
e–folding time of the late starburst population model (Myr)	$\tau_{\rm burst}$	1, 5, 10, 50, 100
Age of the late burst (Myr)	$t_{\rm burst}$	1, 5, 10, 25, 50
Mass fraction of the late burst population	$f_{\rm burst}$	Uniform(0,1) steps of 0.1
Stellar Population Synthesis Model (Bruzual & Charlot 2003)		
Initial mass function	IMF	Chabrier (2003)
Stellar Metallicity ($Z_\odot$)	$Z_\star$	0.0001, 0.0004, 0.004, 0.008, 0.02
Nebular Emission Line Spectrum (CLOUDY v13.01; Ferland et al. 1998, 2013)		
Ionisation parameter	$\log_{10} U$	Uniform(-3,-1) steps of 0.1
Gas Metallicity ($Z_\odot$)	Z_g	0.001 (fixed)
Electron Density (cm^{-3})	n_e	100, 1000
Fraction of Lyman Continuum photons escaping the galaxy	f_{esc}	0.0
Fraction of Lyman Continuum photons absorbed by dust	$f_{esc,\rm dust}$	0.0
Emission Line Widths (km s^{-1})	Δv_{lines}	370 (fixed)
Dust Attenuation Model (Calzetti et al. 2000)		
Reddening of the nebular lines light for young & old population (mag)	$E(B-V)_l$	Uniform(0,0.14) steps of 0.02
Ratio of $E(B-V)_l$ to Stellar Continuum Reddening (mag)	$f_{E(B-V)}$	1.0 (fixed)
Central wavelength of the UV bump (Å)	$\lambda_{UV,b}$	2175 (fixed)
Width (FWHM) of the UV bump (Å)	$\Delta\lambda_{UV,b}$	350
Amplitude of the UV bump (3: Milky Way)	$I_{UV,b}$	0
Slope delta of the power law modifying the attenuation curve	–	0.0
Extinction law to use for attenuating the emission lines flux	–	SMC (Pei 1992)
Total-to-selective extinction ratio for extinction curve applied to emission lines	$A_V/E(B-V)$	3.16
Dust Emission Model (Draine et al. 2014)		
Mass fraction of PAH	q_{PAH}	2.50
Minimum Radiation Field	U_{min}	1.0
Power law slope $dU/dM \propto U^\alpha$	α	2.0
Fraction illuminated from U_{min} to U_{max}	γ	0.1

APPENDIX

A. SOME EXTRA MATERIAL

Table 7. BAGPIPES parameters used in the SED fitting process

Parameter	Range	Prior
Continuity Non-Parametric Star Formation History Model (Leja et al. 2019)		
Mass Formed in Each Time Bin ($\log_{10} M/\mathrm{M}_{\odot}$)	[1,13]	Uniform
Metallicity ($Z_{\odot}$)	0.065	Fixed
Time Bin Edges (Myr)	0, 3, 10, 30, 100, 300, 1000, 3000, 6000	–
Stellar Population Synthesis Model (BPASS v2.2.1 Stanway & Eldridge 2018)		
Initial mass function – Broken Power Law	IMF	–
Nebular Emission Line Spectrum (CLOUDY v17.03)		
Ionisation parameter ($\log_{10} U$)	[-4,-1]	Uniform
Dust Attenuation Model (Calzetti et al. 2000)		
V-band dust attenuation (A_V; mag)	[0,10]	Uniform
$E(B-V)_{nebular}/E(B-V)_{stellar}$	1.0	Fixed
Slope delta of the power law modifying the attenuation curve	0.0	Fixed
Dust Emission Model (Draine et al. 2014)		
Mass fraction of PAH (q_{PAH})	2.50	Fixed
Minimum Radiation Field (U_{min})	1.0	Fixed
Fraction illuminated from U_{min} to U_{max} (γ)	0.1	Fixed